The Interaction Estimand as a Difference in Band Width: A Tutorial

2026-11-11 09:19 PST

Author. pmsimstats team

Audience. Undergraduate statistics students who have taken a course in probability and one in linear regression. You should know what a mean, a variance, a correlation, a regression slope and a t-test are. You do not need to know mixed models. Every other idea is introduced where it is needed.

Purpose. This tutorial expands Section 5 of `docs/40` (the Hendrickson Hybrid design primer). That section shows that the biomarker-by-drug interaction, the quantity a predictive-biomarker trial tries to detect, can be drawn as a band: the gap between the expected outcomes of participants with high and with low biomarker values. The interaction is how much wider the band is when participants are on drug than when they are off drug. Here we derive that statement from the multivariate normal distribution, work it through every path of the design, and follow it to the power of a test. Figure 5 and Table 9 of `docs/40` are expanded into Figures 4 and 5 and Tables 3 to 5 below.

How the numbers were obtained. Every number in this tutorial was computed by the script `03-band-width-tutorial.R` in `analysis/scripts/quick-sim/hybrid-design-primer/` (Section 15) from the published construct (compound symmetry within components, $\rho_{cs} = 0.8$, the parameters of Hendrickson's code at commit `58b32a9`) with carryover anchored at the last on-drug mean (`docs/53`). The script stops unless its closed-form values reproduce Table 9 of `docs/40` and the closed-form E9 power of `docs/37`, Section 5, and it checks the closed forms against 5,000 simulated trials per scenario (Section 11). The study is exploratory.

Contents

Notation index and glossary . . . 4
Glossary . . . 4
1. Summary . . . 5
2. The question and the design . . . 5
3. One visit: the bivariate normal . . . 5
4. Many visits: the band on the trajectory . . . 7
5. From the band to one number: the E9 contrast . . . 8
6. The three ways to set the off-drug band . . . 9
7. Figure 5 of `docs/40`, for every path . . . 9
8. Table 9 of `docs/40`, expanded . . . 10
9. The half-life . . . 11
10. What the band looks like in data . . . 14
11. From the expected slope to power . . . 14
12. Common misreadings . . . 17
13. Exercises . . . 18
14. Scope and limitations . . . 18
15. Reproducibility . . . 18
16. References . . . 19

List of Tables

1 Notation index . . . 4
2 On-drug and off-drug visits of the four Hybrid paths . . . 5
3 Band width W_t (points) at each visit of path 1 under the four scenarios, and the resulting slope . . . 9
4 Per-path and pooled E9 quantities by scenario ($c_{bm} = 0.25$; under mean moderation $\bar{g}$ is the on-drug indicator) . . . 10
5 On/off correlation difference $\mathcal{G}$ by path and half-life under the decaying rule, with the pooled slope, the approximate sample-size factor and the closed-form E9 power . . . 13
6 Variance of the E9 contrast on path 1, compound-symmetry construct . . . 15
7 Closed form against simulation, pooled E9, Hybrid, $N = 70$, 5,000 trials per scenario . . . 17

List of Figures

1 A single visit as a regression of the drug response on the biomarker (path 1, decaying rule, half-life 1 week). The band width is twice the regression slope: 4 points on drug, 1 point two weeks after stopping. . . . 6

2 Expected improvement on path 1 at one SD above, at, and one SD below the biomarker mean, under the four scenarios. The shaded band is the difference the biomarker makes at each visit. 7

3 The correlation profile at an off-drug visit under the decaying rule, by half-life, and under the constant rule. Gray bands mark the Hybrid off-drug visits. 10

4 Band width by visit, path and scenario. Each panel's slope is $-(\bar{W}_{\text{on}} - \bar{W}_{\text{off}})/(2\sigma_{bm})$. 12

5 The on/off correlation difference by half-life under the decaying rule, by path and pooled, with the constant rule (0) and mean moderation (1). 13

6 On-drug and off-drug mean improvement against the biomarker on path 1, with their regression lines. The interaction is the difference between the two slopes. 14

7 The E9 slope in 5,000 simulated trials of $N = 70$ per scenario, with the normal distribution implied by the closed form. 16

8 Closed-form E9 power by half-life under the three rules, with the simulated points. 17

Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here only.

Table 1: Notation index

Symbol	Meaning	Status
i, t, q	participant, visit and path indices	canonical
N, n_q	total sample size (70) and the number on path q (18, 18, 17, 17)	canonical
ϖ_q	allocation share n_q/N	local
Y_{it}	symptom score; lower is better	canonical
BL_i	baseline symptom score	canonical
BR_{it}, PB_{it}, TV_{it}	drug, expectancy and natural-course responses (improvements)	canonical
B_i	biomarker, standing systolic blood pressure (mmHg)	canonical
b_i	standardized biomarker, $(B_i - \mu_{bm})/\sigma_{bm}$	canonical
μ_{bm}, σ_{bm}	biomarker mean (124.33) and SD (15.36)	canonical
σ_{BR}	SD of the drug response (8)	canonical
c_{bm}	biomarker-response correlation on drug (0.25 throughout)	canonical
β_{bm}	mean-moderation strength	canonical
g_t	correlation profile: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\, g_t$	canonical
t_{sd}	weeks since the drug was last stopped	canonical
$t_{1/2}$	carryover half-life (weeks)	canonical
W_t	band width at visit t	local
$\bar{g}_{\mathrm{on}}$, $\bar{g}_{\mathrm{off}}$	mean of g_t over a path's on-drug and off-drug visits	canonical
$\mathcal{G}$	on/off correlation difference, $\bar{g}_{\mathrm{on}} - \bar{g}_{\mathrm{off}}$	canonical
Δ_i	E9 contrast: mean Y on drug minus mean Y off drug	canonical
$\beta_{bm:D}$	interaction slope, the estimand	canonical
V_q, $\sigma^2_{\Delta,q}$	variance of Δ on path q, and its variance given B	local

Sign convention. The three responses are improvements and are subtracted from baseline, $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$. A larger drug response lowers Y. Interaction slopes are therefore negative when higher biomarker values predict more benefit.

Glossary

- **Path.** One sequence of drug states through the trial. The Hybrid design has four.
- **Band.** At one visit, the vertical gap between the expected improvement of a participant one SD above the biomarker mean and that of a participant one SD below it.
- **Correlation profile g_t.** The share of the full biomarker-response correlation c_{bm} present at visit t. NOTATION.md calls it the coupling profile.
- **Constant rule.** After the first exposure, $g_t = 1$ at every visit at which the mean drug response is not zero. It is the rule in Hendrickson's code (NOTATION.md: step coupling).
- **Decaying rule.** $g_t = 1$ on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before any exposure. It is the rule of paper 01's recommended data-generating process (NOTATION.md: graded coupling).
- **Covariance moderation and mean moderation.** Two ways to build the interaction into simulated data. Covariance moderation sets the correlation between B and BR (both rules above). Mean moderation adds $\beta_{bm}\sigma_{BR}b_i$ to the drug response at on-drug visits.
- **E9.** The paired-difference statistic of `docs/37`: each participant's on-drug mean minus off-drug mean of Y, regressed on the biomarker. The baseline visit is not used.
- **Power.** The probability that a test rejects the null hypothesis of no interaction when an interaction is present.

1. Summary

1. **At one visit the interaction is a regression slope.** For a bivariate normal pair, the conditional mean of one variable is a straight line in the other. The drug response at visit t rises by $c_{bm} g_t \sigma_{BR}$ points per SD of biomarker, so the band at that visit is $W_t = 2c_{bm}\sigma_{BR} g_t$ points wide: 4 points at full correlation (Section 3, Figure 1).
2. **Only the drug response moves with the biomarker.** The expectancy and natural-course responses and baseline are uncorrelated with the biomarker, so they shift the band up and down but do not change its width (Section 4, Figure 2).
3. **The estimand is a difference in band width.** The E9 slope has expectation $\beta_{bm:D} = -(\bar{W}_{\text{on}} - \bar{W}_{\text{off}})/(2\sigma_{bm}) = -0.130\,\mathcal{G}$ points per mmHg (Section 5).
4. **The correlation rule decides the off-drug band.** Under the constant rule any carryover keeps the band fully open off drug, so $\mathcal{G} = 0$ and the estimand is exactly zero. Under the decaying rule the band narrows as the drug effect fades; at a one-week half-life $\mathcal{G}$ is 0.73 to 0.78 by path (Sections 6 to 8, Figure 4, Table 4).
5. $\mathcal{G}$ **falls smoothly and strictly with the half-life** under the decaying rule, from 1 toward 0. The sample size needed for a given power grows roughly as $1/\mathcal{G}^2$: about 1.8 times at a one-week half-life and 3.3 times at two weeks (Section 9, Figure 5, Table 5).
6. **The same expected slope does not mean the same power.** Under covariance moderation the biomarker explains part of the existing variation of the contrast. Under mean moderation it adds new variation and then explains it. At the same expected slope, −0.130, power at $N = 70$ is 0.721 and 0.682 (Section 11, Table 6).

2. The question and the design

Hendrickson et al. [1] planned a trial of prazosin for posttraumatic stress disorder. The question was whether standing systolic blood pressure at baseline predicts how much a participant improves on prazosin. A predictor of the size of a treatment effect is a predictive biomarker, and the quantity that measures it is a biomarker-by-treatment interaction.

The trial used the Hybrid design (`docs/40`, Section 2): 8 weeks of open-label drug, a 4-week blinded discontinuation block in which the drug stops after week 9 or after week 10, and a blinded two-period crossover. There are eight post-baseline visits, at weeks 4, 8, 9, 10, 11, 12, 16 and 20. Two randomizations give four paths (Table 2).

Table 2: On-drug and off-drug visits of the four Hybrid paths

Path	n_q	On drug at weeks	Off drug at weeks (t_{sd} in weeks)
1	18	4, 8, 9, 10, 16	11 (1), 12 (2), 20 (4)
2	18	4, 8, 9, 10, 20	11 (1), 12 (2), 16 (6)
3	17	4, 8, 9, 16	10 (1), 11 (2), 12 (3), 20 (4)
4	17	4, 8, 9, 20	10 (1), 11 (2), 12 (3), 16 (7)

Every participant is observed both on and off drug. Each participant therefore supplies a within-person comparison, and the interaction becomes a question about how those comparisons vary with the biomarker. The rest of this tutorial makes that sentence exact.

3. One visit: the bivariate normal

Fix one visit t and consider two quantities for a participant: the standardized biomarker b and the drug response BR_t. In the simulation they are jointly normal, with

$$\mathrm{E}(b) = 0, \quad \mathrm{Var}(b) = 1, \quad \mathrm{E}(BR_t) = \mu_t, \quad \mathrm{Var}(BR_t) = \sigma^2_{BR}, \quad \mathrm{Cor}(b, BR_t) = c_{bm}\, g_t.$$

The conditional mean is a straight line. For any jointly normal pair (X, Y),

$$\mathrm{E}(Y \mid X = x) = \mu_Y + \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}\,(x - \mu_X).$$

The proof takes three lines. Let $\gamma = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$ and define the residual $e = Y - \mu_Y - \gamma(X - \mu_X)$. Then $\mathrm{Cov}(e, X) = \mathrm{Cov}(Y, X) - \gamma\,\mathrm{Var}(X) = 0$. Because e and X are jointly normal, zero covariance implies independence, so $\mathrm{E}(e \mid X) = \mathrm{E}(e) = 0$, which is the result. (Without normality the same line is the best *linear* predictor, but it need not be the conditional mean.)

Applied to our pair, $\mathrm{Cov}(b, BR_t) = c_{bm} g_t \sigma_{BR}$ and $\mathrm{Var}(b) = 1$, so

$$\mathrm{E}(BR_t \mid b) = \mu_t + c_{bm}\, g_t\, \sigma_{BR}\, b.$$

The slope $c_{bm} g_t \sigma_{BR}$ is in points per SD of biomarker. A participant one SD above the mean expects $c_{bm} g_t \sigma_{BR}$ more improvement than an average participant, and one SD below expects that much less. The **band width** at visit t is the distance between them:

$$W_t = \mathrm{E}(BR_t \mid b = 1) - \mathrm{E}(BR_t \mid b = -1) = 2\, c_{bm}\, \sigma_{BR}\, g_t.$$

With $c_{bm} = 0.25$ and $\sigma_{BR} = 8$, the full width ($g_t = 1$) is $2 \times 0.25 \times 8 = 4$ points.

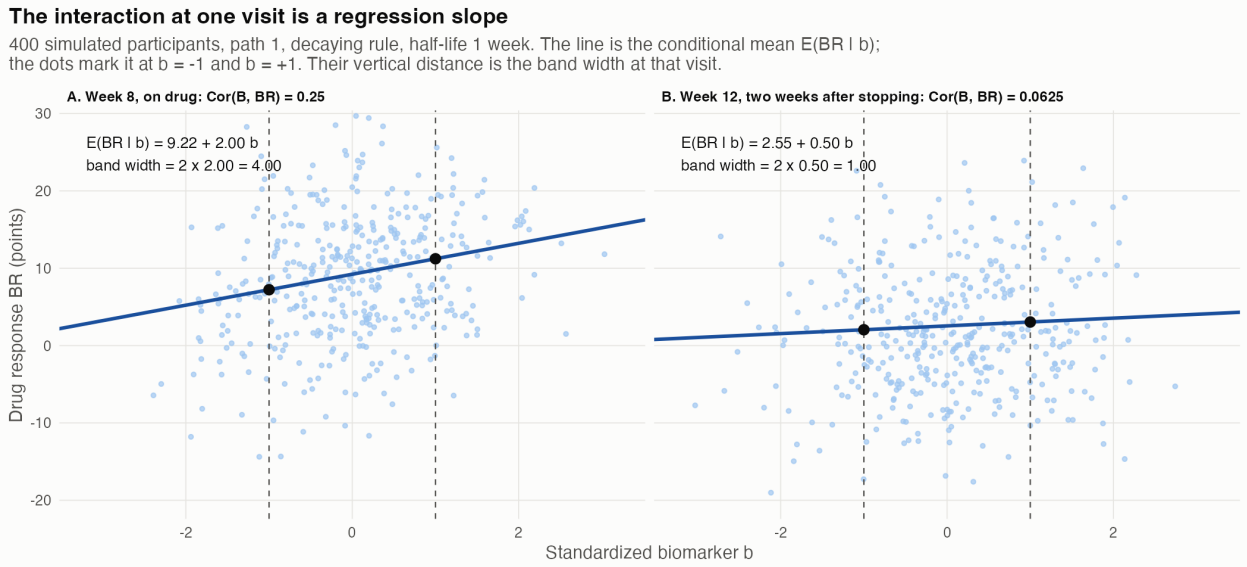


Figure 1: A single visit as a regression of the drug response on the biomarker (path 1, decaying rule, half-life 1 week). The band width is twice the regression slope: 4 points on drug, 1 point two weeks after stopping.

Figure 1 shows this for two visits of path 1 under the decaying rule with a one-week half-life. At week 8 the participant is on drug and $g_t = 1$: the line rises 2 points per SD and the band is 4 points wide. At week 12 the drug stopped two weeks earlier, $g_t = 2^{-2} = 0.25$, the correlation is $0.25 \times 0.25 = 0.0625$, and the band is 1 point wide. The scatter around each line is wide. A correlation of 0.25 explains only $0.25^2 = 6\%$ of the variance of the drug response at a visit. The interaction is a small signal, which is why its power is a question at all.

4. Many visits: the band on the trajectory

A participant contributes 26 jointly normal quantities: the biomarker, baseline, and TV, PB and BR at each of the eight visits. Their correlations with each other are many (`docs/40`, Section 6), but the conditional mean of each one given b depends only on its own covariance with b. This follows from the multivariate normal: for a normal vector $(X, Y_1, \dots, Y_k)$, each $\mathrm{E}(Y_j \mid X)$ is given by the bivariate formula of Section 3 applied to the pair (X, Y_j). The correlations among the Y_j matter for variances, which Section 11 takes up, and not for conditional means.

In the published construct the biomarker is correlated with BR_t and with nothing else. Baseline, TV_t and PB_t have zero covariance with b, so their conditional means equal their unconditional means. The expected total improvement at visit t is

$$\mathrm{E}(TV_t + PB_t + BR_t \mid b) = \underbrace{\mathrm{E}(TV_t) + \mathrm{E}(PB_t) + \mu_t}_{\text{the trajectory at } b=0} + c_{bm}\, g_t\, \sigma_{BR}\, b.$$

Only the last term depends on the biomarker. Expectancy and natural course raise and lower the whole band together, visit by visit; they do not change its width.

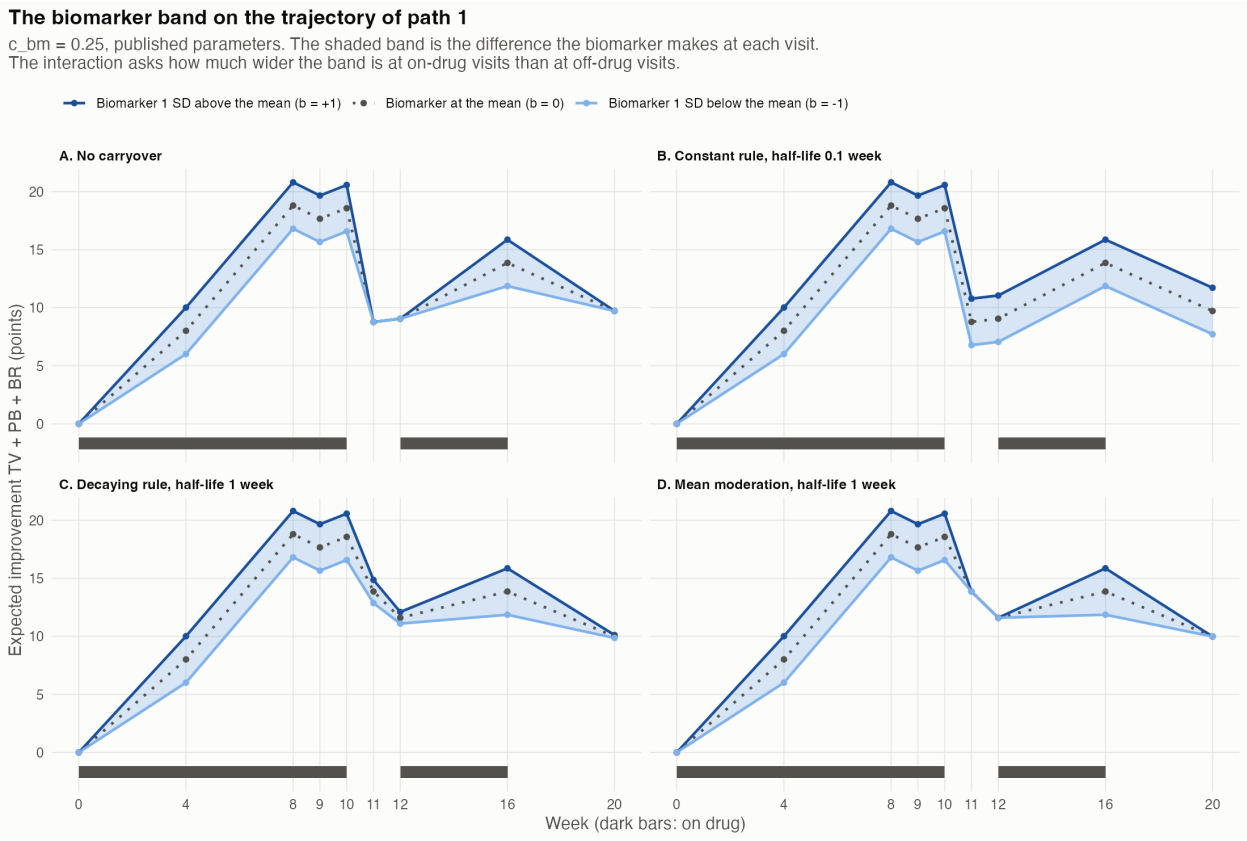


Figure 2: Expected improvement on path 1 at one SD above, at, and one SD below the biomarker mean, under the four scenarios. The shaded band is the difference the biomarker makes at each visit.

Figure 2 draws the band for path 1 under four scenarios used throughout:

- **A. No carryover.** The band is 4 points wide at the five on-drug visits and closes completely at the three off-drug visits.

- **B. Constant rule, half-life 0.1 week.** The mean drug response after stopping is tiny (0.1% of its last on-drug value after one week), but it is not zero. The constant rule therefore keeps $g_t = 1$, and the band stays fully open off drug.
- **C. Decaying rule, half-life 1 week.** The band narrows after the drug stops: 2 points one week later, 1 point after two weeks and 0.25 point after four.
- **D. Mean moderation, half-life 1 week.** The shift $\beta_{bm}\sigma_{BR}b$ is added only at on-drug visits. The band closes as soon as the drug stops, whatever the half-life, although the mean drug response still decays slowly (the middle line).

In every panel the on-drug band is the same. The scenarios differ only off drug. That is the whole story of the interaction under carryover.

5. From the band to one number: the E9 contrast

A test needs one number per participant. The E9 statistic of `docs/37` takes, for participant i on path q,

$$\Delta_i = \frac{1}{n_{\text{on}}} \sum_{t\,\text{on}} Y_{it} \; - \; \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} Y_{it},$$

the mean symptom score on drug minus the mean off drug over the eight post-baseline visits, and regresses Δ_i on B_i. It is a two-stage analysis of the kind taught as summary measures: reduce each participant's repeated measurements to one contrast, then regress the contrasts on a covariate.

The expected contrast. Baseline cancels, because it appears in both means with weight 1. By linearity of expectation and Section 4,

$$\mathrm{E}(\Delta_i \mid b_i) = \text{const} - c_{bm}\,\sigma_{BR}\,(\bar{g}_{\text{on}} - \bar{g}_{\text{off}})\,b_i,$$

where the constant collects the expected TV, PB and BR terms and $\bar{g}_{\text{on}}$, $\bar{g}_{\text{off}}$ are the means of g_t over the path's on-drug and off-drug visits. Converting to the raw biomarker with $b_i = (B_i - \mu_{bm})/\sigma_{bm}$ gives the slope of $\mathrm{E}(\Delta \mid B)$ in points per mmHg:

$$\beta_{bm:D} = -\,c_{bm}\,\frac{\sigma_{BR}}{\sigma_{bm}}\,\mathcal{G}, \qquad \mathcal{G} = \bar{g}_{\text{on}} - \bar{g}_{\text{off}}.$$

The same slope, written with the band. Since $W_t = 2c_{bm}\sigma_{BR}g_t$, the mean band widths are $\bar{W}_{\text{on}} = 2c_{bm}\sigma_{BR}\bar{g}_{\text{on}}$ and $\bar{W}_{\text{off}} = 2c_{bm}\sigma_{BR}\bar{g}_{\text{off}}$, and

$$\beta_{bm:D} = -\,\frac{\bar{W}_{\text{on}} - \bar{W}_{\text{off}}}{2\,\sigma_{bm}}.$$

The numerator is how much wider the band is on drug than off drug. The denominator converts "per 2 SD of biomarker" (the band compares $b = +1$ with $b = -1$) into "per mmHg". This is the statement of `docs/40`, Section 5, now derived.

The numbers. With the parameters above, $c_{bm}\sigma_{BR}/\sigma_{bm} = 0.25 \times 8/15.36 = 0.130$, so $\beta_{bm:D} = -0.130\,\mathcal{G}$ points per mmHg. Without carryover, $\mathcal{G} = 1$: a participant whose blood pressure is 10 mmHg higher has an on-minus-off contrast 1.30 points lower, that is, 1.30 more points of improvement on drug relative to off drug.

Table 3 works path 1 through visit by visit.

Table 3: Band width W_t (points) at each visit of path 1 under the four scenarios, and the resulting slope

Week	State	t_{sd}	A: none	B: constant, 0.1	C: decaying, 1	D: mean, 1
4	on	0	4	4	4	4
8	on	0	4	4	4	4
9	on	0	4	4	4	4
10	on	0	4	4	4	4
11	off	1	0	4	2	0
12	off	2	0	4	1	0
16	on	0	4	4	4	4
20	off	4	0	4	0.25	0
$\bar{W}_{\text{on}}$			4	4	4	4
$\bar{W}_{\text{off}}$			0	4	1.083	0
$\bar{W}_{\text{on}} - \bar{W}_{\text{off}}$			4	0	2.917	4
$\mathcal{G}$			1	0	0.729	1
$\beta_{bm:D}$			-0.130	0	-0.095	-0.130

For scenario C, for example, $\bar{W}_{\text{off}} = (2 + 1 + 0.25)/3 = 1.083$, the difference is $4 - 1.083 = 2.917$, and $\beta_{bm:D} = -2.917/(2 \times 15.36) = -0.095$.

6. The three ways to set the off-drug band

The on-drug band is the same in every scenario: $g_t = 1$ on drug. All the action is in g_t off drug.

Decaying rule. $g_t = 2^{-t_{sd}/t_{1/2}}$ off drug after exposure. The biomarker predicts the size of the drug effect, so as the effect fades the biomarker should predict less (paper 01, Stage 2). Figure 3 shows the profile for five half-lives, with the values of t_{sd} at which the Hybrid paths actually have off-drug visits. Because every off-drug visit is at least one week after stopping, a short half-life leaves almost nothing: at $t_{1/2} = 0.1$ week, $g_t \leq 2^{-10} < 0.001$.

Constant rule. $g_t = 1$ wherever the mean drug response is not zero. With any positive half-life the mean is positive at every visit after the first exposure, so $g_t = 1$ at every off-drug visit, and $\bar{g}_{\text{off}} = 1 = \bar{g}_{\text{on}}$. The band is equally wide on and off drug, and $\mathcal{G} = 0$ exactly. The constant rule is the decaying rule in the limit $t_{1/2} \to \infty$, applied at every positive half-life. It therefore jumps: $\mathcal{G} = 1$ at $t_{1/2} = 0$ and $\mathcal{G} = 0$ at $t_{1/2} = 0.01$. (Below about 0.007 week the mean drug response underflows to exactly zero in double-precision arithmetic, and the constant rule then behaves as if there were no carryover; this is an artifact of the computer, not of the rule; `docs/37`.)

Mean moderation. Here the biomarker shifts the drug response directly, $BR_t + \beta_{bm}\sigma_{BR}b$ at on-drug visits, and the correlation row stays zero. The band is $2\beta_{bm}\sigma_{BR}$ wide on drug and closed off drug at every half-life, so $\mathcal{G} = 1$ and the slope is $-\beta_{bm}\sigma_{BR}/\sigma_{bm}$. At $\beta_{bm} = 0.25$ it equals the no-carryover slope of covariance moderation, -0.130. Mean moderation describes a biomarker that acts only while the drug is taken, even if the drug's effect on the mean lingers. Paper 01 shows that, conditional on the biomarker, covariance moderation is a mean-term form with a smaller residual variance (its Discussion, "The correlation encoding of the interaction"), and it does not claim either form is preferable. Section 11 shows the variance difference at work.

7. Figure 5 of `docs/40`, for every path

Figure 5 of `docs/40` showed the band width visit by visit for path 1. Figure 4 repeats it for all four paths. Rows are the four scenarios; columns are the paths. Blue bars are on-drug visits, gray bars off-drug visits; the solid and dashed lines are their means; the text gives $\mathcal{G}$ and $\beta_{bm:D}$.

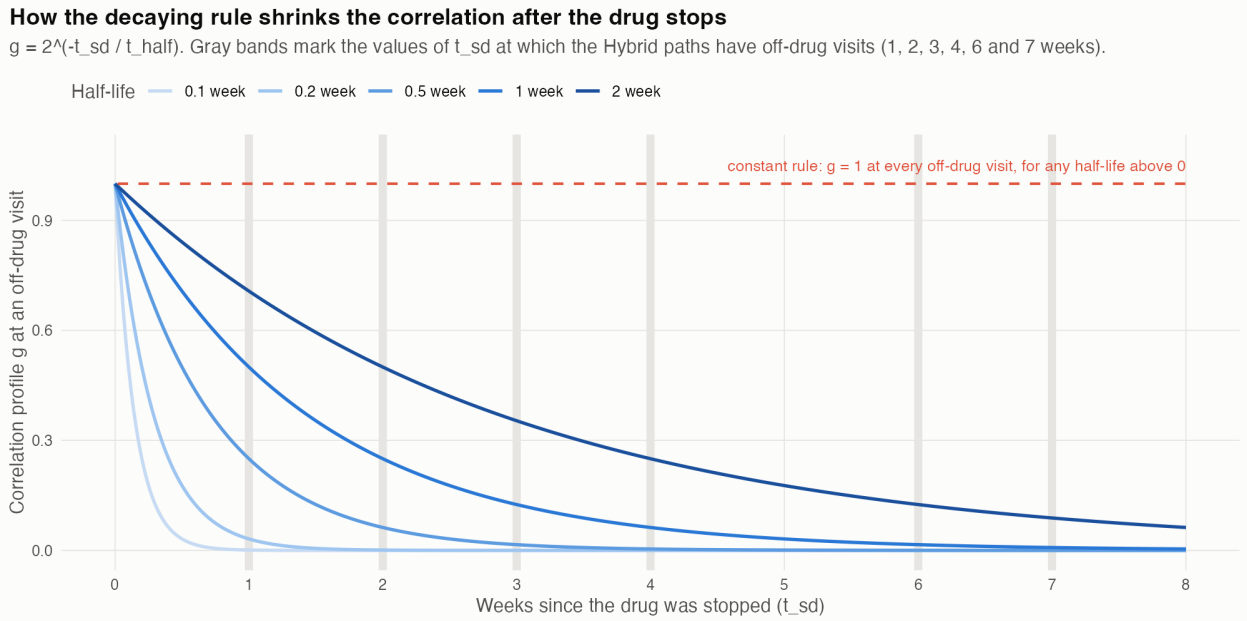


Figure 3: The correlation profile at an off-drug visit under the decaying rule, by half-life, and under the constant rule. Gray bands mark the Hybrid off-drug visits.

How to read it:

- **Rows A and D** look alike: blue bars at 4, no gray bars, so $\mathcal{G} = 1$ in every path. They differ in how the data are generated (Section 11), not in the expected slope.
- **Row B** has gray bars as tall as the blue ones in every path. The solid and dashed lines coincide at 4, and the slope is zero. No analysis can detect an interaction that is not there, which is why the power that Hendrickson et al. reported under carryover collapsed (`docs/37`, `docs/55`).
- **Row C** is the informative one. The gray bars step down after each discontinuation: 2, 1, 0.5, 0.25 points one to four weeks after stopping. A path whose off-drug visits lie further from a discontinuation has shorter gray bars, a lower dashed line and a larger $\mathcal{G}$.

8. Table 9 of `docs/40`, expanded

Table 9 of `docs/40` listed the per-path slopes. Table 4 adds the quantities that produce them and the pooled value.

Table 4: Per-path and pooled E9 quantities by scenario ($c_{bm} = 0.25$; under mean moderation $\bar{g}$ is the on-drug indicator)

Scenario	Quantity	Path 1	Path 2	Path 3	Path 4	Pooled
A. No carryover	$\bar{g}_{\text{off}}$	0	0	0	0	
	$\mathcal{G}$	1	1	1	1	1
	$\beta_{bm:D}$	−0.130	−0.130	−0.130	−0.130	−0.130
B. Constant rule, 0.1 week	$\bar{g}_{\text{off}}$	1	1	1	1	
	$\mathcal{G}$	0	0	0	0	0
	$\beta_{bm:D}$	0	0	0	0	0
C. Decaying rule, 1 week	$\bar{g}_{\text{off}}$	0.271	0.255	0.234	0.221	

Scenario	Quantity	Path 1	Path 2	Path 3	Path 4	Pooled
	$\mathcal{G}$	0.729	0.745	0.766	0.779	0.754
	$\beta_{bm:D}$	−0.095	−0.097	−0.100	−0.101	−0.098
D. Mean moderation, 1 week	$\bar{g}_{\text{off}}$	0	0	0	0	
	$\mathcal{G}$	1	1	1	1	1
	$\beta_{bm:D}$	−0.130	−0.130	−0.130	−0.130	−0.130

A worked example: path 4, scenario C. The off-drug visits are at $t_{sd} = 1, 2, 3, 7$ weeks (Table 2). With $t_{1/2} = 1$,

$$\bar{g}_{\text{off}} = \tfrac{1}{4}(2^{-1} + 2^{-2} + 2^{-3} + 2^{-7}) = \tfrac{1}{4}(0.5 + 0.25 + 0.125 + 0.0078) = 0.221,$$

so $\mathcal{G} = 0.779$ and $\beta_{bm:D} = -0.130 \times 0.779 = -0.101$. Path 1 has three off-drug visits at $t_{sd} = 1, 2, 4$, giving $\bar{g}_{\text{off}} = (0.5 + 0.25 + 0.0625)/3 = 0.271$. Path 4 does better because its fourth off-drug visit, seven weeks after stopping, contributes almost nothing to the off-drug band and pulls the average down.

Why the pooled slope is a weighted average. E9 fits one line through all 70 participants with a common intercept. Suppose, within path q, $\mathrm{E}(\Delta \mid B, q) = \mu_q + \beta_q(B - \mu_{bm})$, and the biomarker has the same distribution on every path, as it does under randomization. By the law of total covariance,

$$\mathrm{Cov}(\Delta, B) = \sum_q \varpi_q \,\mathrm{Cov}(\Delta, B \mid q) + \mathrm{Cov}(\mathrm{E}(\Delta \mid q), \mathrm{E}(B \mid q)) = \sum_q \varpi_q \,\beta_q \,\sigma^2_{bm} + 0,$$

because $\mathrm{E}(B \mid q) = \mu_{bm}$ on every path. Dividing by $\mathrm{Var}(B) = \sigma^2_{bm}$, the pooled slope converges to $\bar{\beta} = \sum_q \varpi_q \beta_q$. For scenario C, $(18 \times 0.0949 + 18 \times 0.0970 + 17 \times 0.0997 + 17 \times 0.1015)/70 = 0.0982$, with a minus sign. The differences between the path means μ_q do not bias the slope; they add to its variance (Section 11).

9. The half-life

Under the decaying rule, $\bar{g}_{\text{off}}$ on path q is an explicit function of the half-life:

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}}, \qquad \mathcal{G}(t_{1/2}) = 1 - \bar{g}_{\text{off}}(t_{1/2}).$$

Three properties follow from calculus.

1. *Strictly decreasing.* Each term has derivative $\frac{d}{dt_{1/2}} 2^{-s/t_{1/2}} = 2^{-s/t_{1/2}} \ln 2 \, s/t^2_{1/2} > 0$ for $s > 0$, so $\bar{g}_{\text{off}}$ increases and $\mathcal{G}$ decreases strictly with the half-life (paper 01, Appendix A.3).
2. *Flat near zero.* Every off-drug visit has $s \geq 1$, so $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$, which goes to zero faster than any power of $t_{1/2}$. At the pharmacological half-lives of prazosin, 0.1 to 0.2 week, $\mathcal{G}$ is within about 0.01 of 1.
3. *Constant rule at infinity.* As $t_{1/2} \to \infty$ every term tends to 1, so $\mathcal{G} \to 0$, the value the constant rule takes at every positive half-life.

Figure 5 plots $\mathcal{G}$ against the half-life on a log scale, and Table 5 tabulates it.

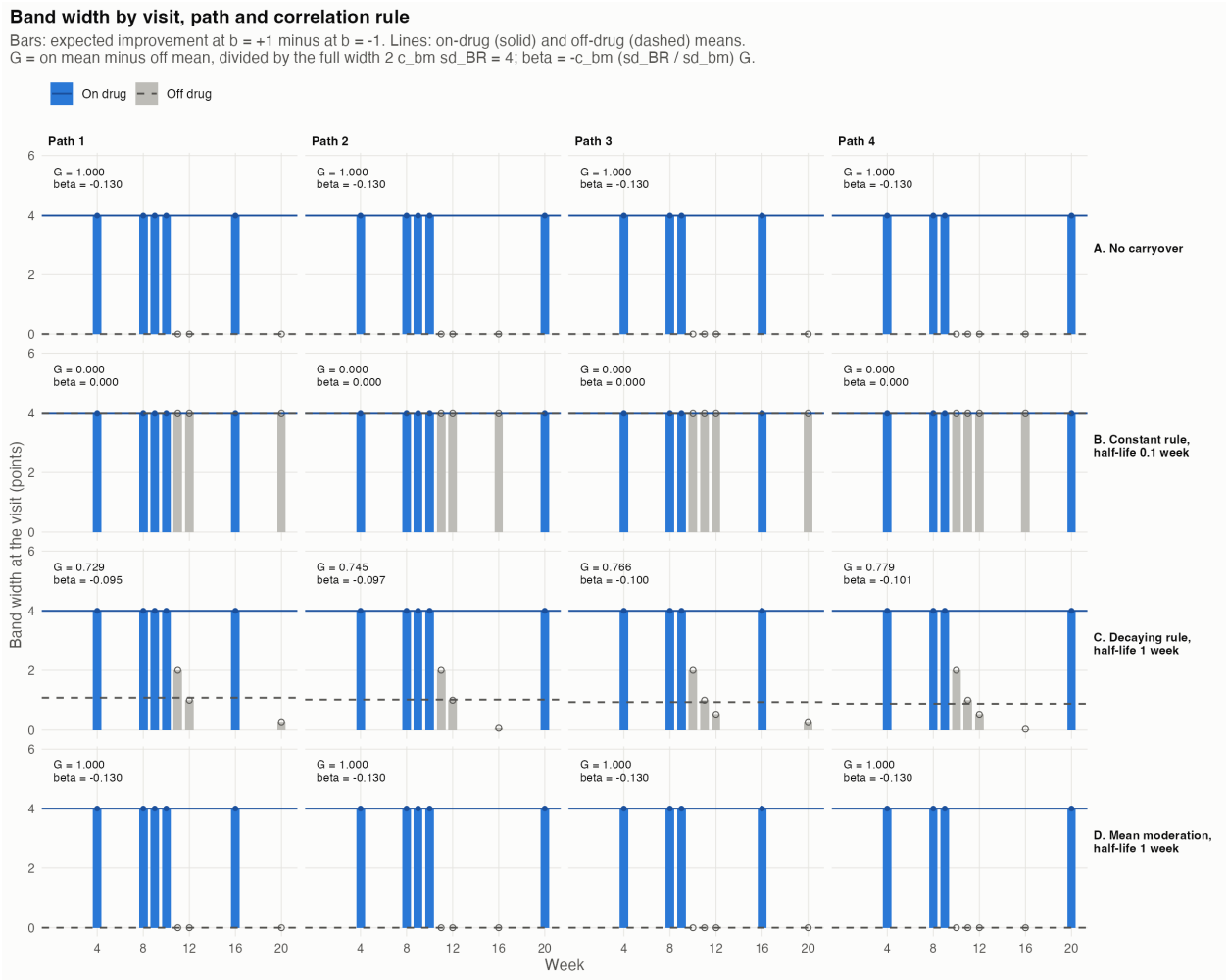


Figure 4: Band width by visit, path and scenario. Each panel's slope is $-(\bar{W}_{\text{on}} - \bar{W}_{\text{off}})/(2\sigma_{bm})$.

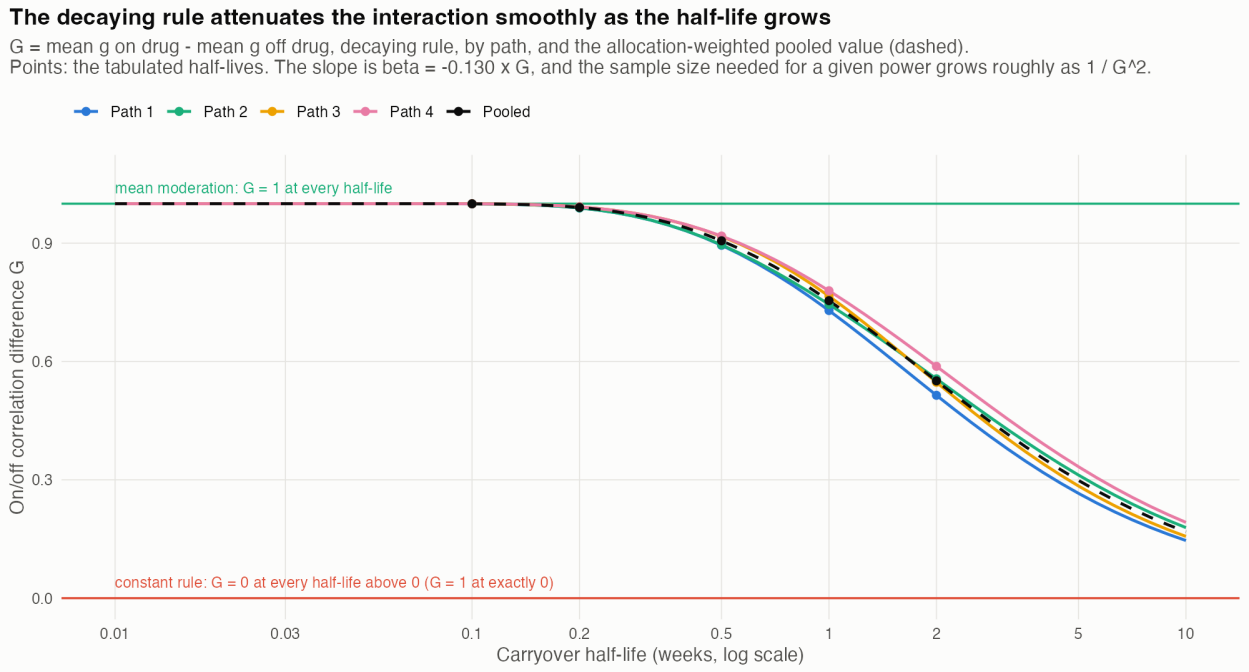


Figure 5: The on/off correlation difference by half-life under the decaying rule, by path and pooled, with the constant rule (0) and mean moderation (1).

Table 5: On/off correlation difference $\mathcal{G}$ by path and half-life under the decaying rule, with the pooled slope, the approximate sample-size factor and the closed-form E9 power

$t_{1/2}$ (weeks)	Path 1	Path 2	Path 3	Path 4	Pooled $\mathcal{G}$	Pooled $\beta_{bm:D}$	$1/\mathcal{G}^2$	Power, $N = 70$
0.1	1.000	1.000	1.000	1.000	1.000	−0.130	1.00	0.721
0.2	0.989	0.989	0.992	0.992	0.991	−0.129	1.02	0.712
0.5	0.895	0.896	0.917	0.918	0.906	−0.118	1.22	0.631
1	0.729	0.745	0.766	0.779	0.754	−0.098	1.76	0.473
2	0.514	0.556	0.547	0.588	0.551	−0.072	3.30	0.278

Why the sample size scales as $1/\mathcal{G}^2$. Section 11 shows that the test statistic is approximately normal with mean $|\bar{\beta}|\,\sigma_{bm}\sqrt{N-3}/\sigma_{\text{res}}$, where σ_{res} is the residual SD of the contrast. Power is fixed when that mean is fixed. Since $\bar{\beta}$ is proportional to $\mathcal{G}$, holding power constant requires $N-3$ proportional to $\sigma_{\text{res}}^2/\mathcal{G}^2$. The residual SD changes little with $\mathcal{G}$ (6.38 to 6.50 points between scenarios A and C), so to a good approximation the sample size scales as $1/\mathcal{G}^2$. A trial planned for 70 participants without carryover needs about $70 \times 1.76 \approx 123$ for the same power at a one-week half-life. The ordering of the paths is not fixed: path 2 overtakes path 3 between one and two weeks. At short half-lives path 3's off-drug visits three and four weeks after stopping carry almost no correlation and dilute its off-drug mean. At longer half-lives they still carry some, while path 2's only late visit, six weeks after stopping, carries little.

10. What the band looks like in data

A band is a statement about expectations. Figure 6 shows how it appears in simulated data. For 20,000 participants on path 1, each participant's mean improvement over the on-drug visits and over the off-drug visits is plotted against the biomarker, with a least-squares line through each cloud.

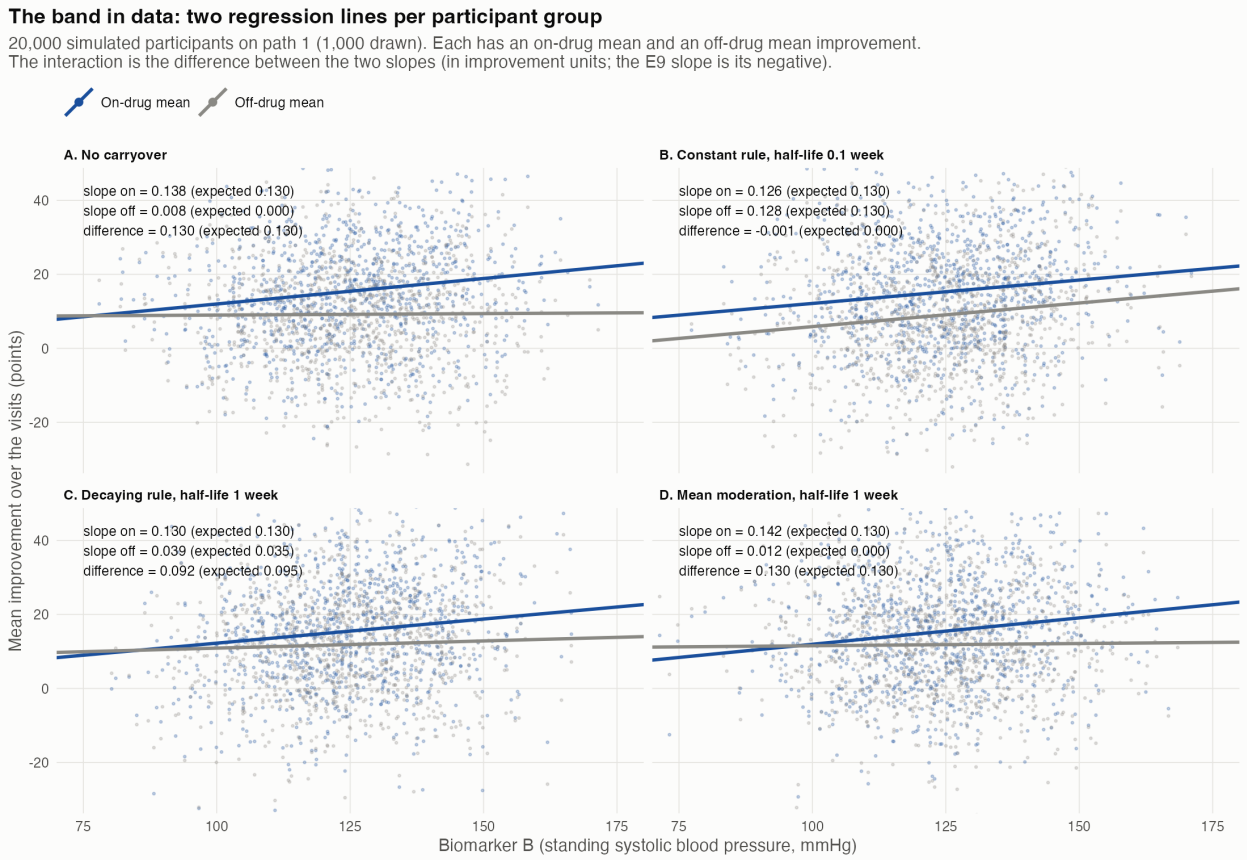


Figure 6: On-drug and off-drug mean improvement against the biomarker on path 1, with their regression lines. The interaction is the difference between the two slopes.

The expected slope of each line is $0.130\,\bar{g}$ points per mmHg: 0.130 on drug in every scenario, and $0.130\,\bar{g}_{\text{off}}$ off drug. The interaction is the **difference between the two slopes**. In panel A the off-drug line is flat and the lines diverge; in panel B they are parallel; in panel C the off-drug line rises at about a quarter of the on-drug slope (expected 0.035 against 0.130). The fitted slopes in the figure agree with their expectations to within sampling error (Monte Carlo SE about 0.006).

The link to a regression with an interaction term. Suppose one fits $Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_{bm:D} B_i D_{it} + \text{error}$, with $D_{it} = 1$ on drug and 0 off. The slope of Y on B is β_B at off-drug visits and $\beta_B + \beta_{bm:D}$ at on-drug visits, so $\beta_{bm:D}$ is the difference between the two slopes of Figure 6, with the sign reversed because Y falls as improvement rises. That is why the published mixed-model analysis and E9 target the same quantity. They weight the visits differently, so their estimates are not identical; `docs/37`, Section 6, compares them.

11. From the expected slope to power

The expected slope says what a test is looking for. Whether it finds it depends also on the noise.

The variance of the contrast. Δ_i is a fixed linear combination $a^\top X_i$ of the participant's 26 normal quantities, so $V_q = \mathrm{Var}(\Delta_i) = a^\top \Sigma a$, with Σ their covariance matrix. Under covariance moderation the biomarker row of Σ does not enter $a^\top \Sigma a$ (the contrast does not contain B), so V_q does not depend on c_{bm}: it is 44.13 on paths 1 and 2 and 45.02 on paths 3 and 4 (compound-symmetry construct). The **residual variance** of the contrast given the biomarker is

$$\sigma^2_{\Delta,q} = V_q - \beta_q^2\,\sigma^2_{bm} = V_q - \left(c_{bm}\sigma_{BR}\mathcal{G}_q\right)^2 = V_q - 4\,\mathcal{G}_q^2.$$

In words, the biomarker explains part of the variation the contrast already has. Under mean moderation the shift adds $-c_{bm}\sigma_{BR}b_i$ to every contrast, so the variance grows to $V_q + 4$, and the biomarker then explains exactly the part it added: the residual variance stays at V_q. Table 6 shows the consequence on path 1.

Table 6: Variance of the E9 contrast on path 1, compound-symmetry construct

Scenario	$\mathrm{Var}(\Delta)$	Slope $\times\,\sigma_{bm}$	Residual variance	$\mathrm{Cor}(\Delta, B)$
A. No carryover	44.13	−2.00	40.13	−0.301
B. Constant rule, 0.1 week	44.13	0	44.13	0
C. Decaying rule, 1 week	44.13	−1.46	42.01	−0.219
D. Mean moderation, 1 week	48.13	−2.00	44.13	−0.288

Scenarios A and D have the same expected slope, but A's residual variance is 4 points smaller. That is the whole reason covariance moderation gives more power than mean moderation at the same slope (0.721 against 0.682).

The sampling distribution of the slope. Conditional on the biomarker values, the least-squares slope is normal with variance $\sigma^2_{\mathrm{res}}/\sum_i (B_i - \bar{B})^2$. Averaging over the biomarker values gives, for the pooled E9 slope (`docs/37`, Section 4.4),

$$\mathrm{SE}(\hat{\beta}) \approx \frac{\sigma_{\mathrm{res}}}{\sigma_{bm}\sqrt{N-3}}, \qquad \sigma^2_{\mathrm{res}} = \sum_q \varpi_q \Big[\sigma^2_{\Delta,q} + (\mu_q - \bar{\mu})^2 + \sigma^2_{bm}(\beta_q - \bar{\beta})^2\Big].$$

The second and third terms are the price of pooling paths with different means and slopes under one intercept; in the Hybrid design they are below 1% of the total.

Power. A two-sided test at level $\alpha = 0.05$ rejects when $|\hat{\beta}|/\widehat{\mathrm{SE}}$ exceeds about 1.96. With $z = |\bar{\beta}|/\mathrm{SE}$,

$$\text{power} \approx \Phi(z - 1.96) + \Phi(-z - 1.96).$$

For scenario A, $\sigma_{\mathrm{res}} = 6.376$, so $\mathrm{SE} = 6.376/(15.36\sqrt{67}) = 0.0507$, $z = 0.130/0.0507 = 2.57$, and power is about $\Phi(0.61) = 0.73$. The exact calculation of `docs/37` replaces the normal by the noncentral t distribution on $N - 2$ degrees of freedom and averages over the biomarker's sum of squares; it gives 0.721.

Figure 7 shows the simulated sampling distributions. Under the constant rule (panel B) the distribution is centered on zero and the test rejects 5% of the time: carryover under that rule turns the trial into a null trial. Table 7 compares the closed forms with the simulation.

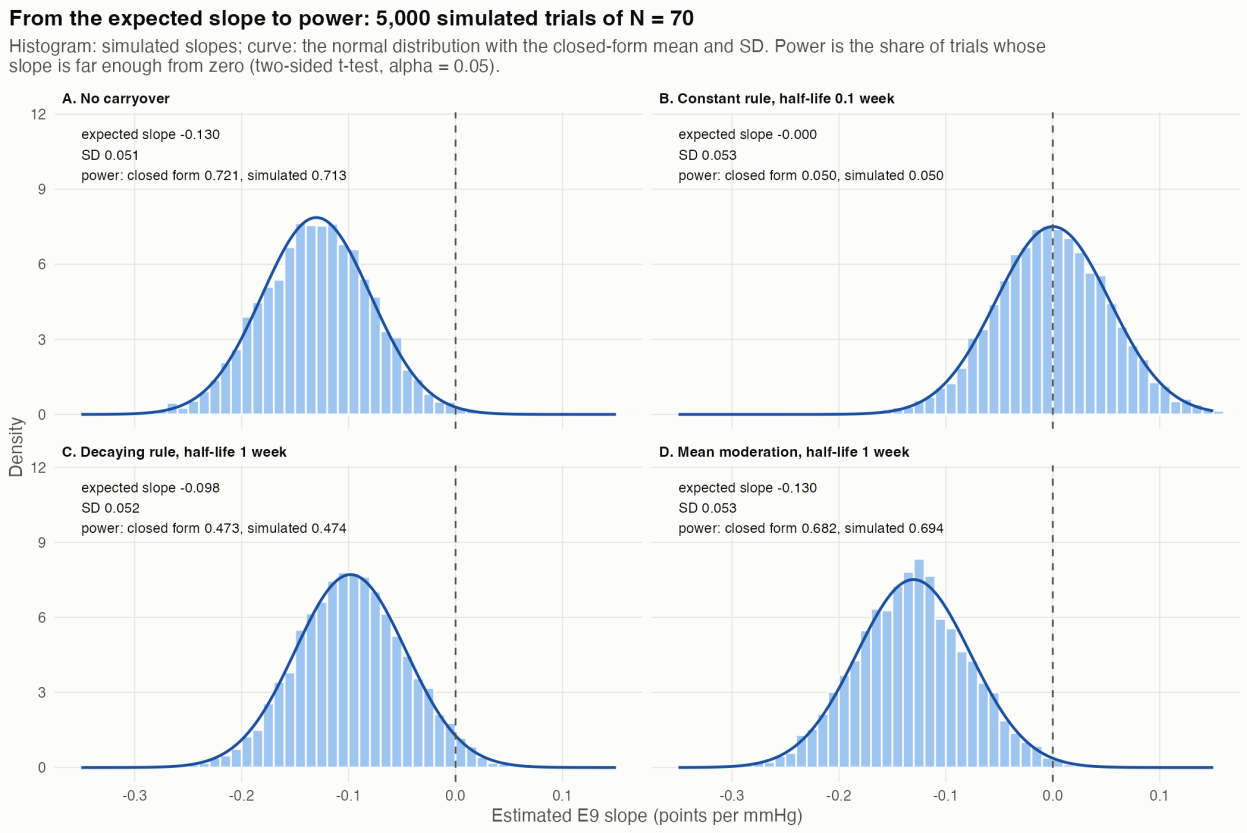


Figure 7: The E9 slope in 5,000 simulated trials of $N = 70$ per scenario, with the normal distribution implied by the closed form.

Table 7: Closed form against simulation, pooled E9, Hybrid, $N = 70$, 5,000 trials per scenario

Scenario	Expected slope	SD, closed form	Simulated mean	Simulated SD	Power, exact	Power, normal approx.	Power, simulated (MCSE)
A. No carryover	−0.1302	0.0507	−0.1297	0.0515	0.721	0.728	0.713 (0.006)
B. Constant rule, 0.1	0	0.0532	−0.0004	0.0535	0.050	0.050	0.050 (0.003)
C. Decaying rule, 1	−0.0982	0.0517	−0.0971	0.0512	0.473	0.476	0.474 (0.007)
D. Mean moderation, 1	−0.1302	0.0531	−0.1314	0.0531	0.682	0.689	0.694 (0.007)

Every simulated mean and power lies within 2 Monte Carlo SE of the closed form (largest discrepancy 1.7 SE, scenario D power).

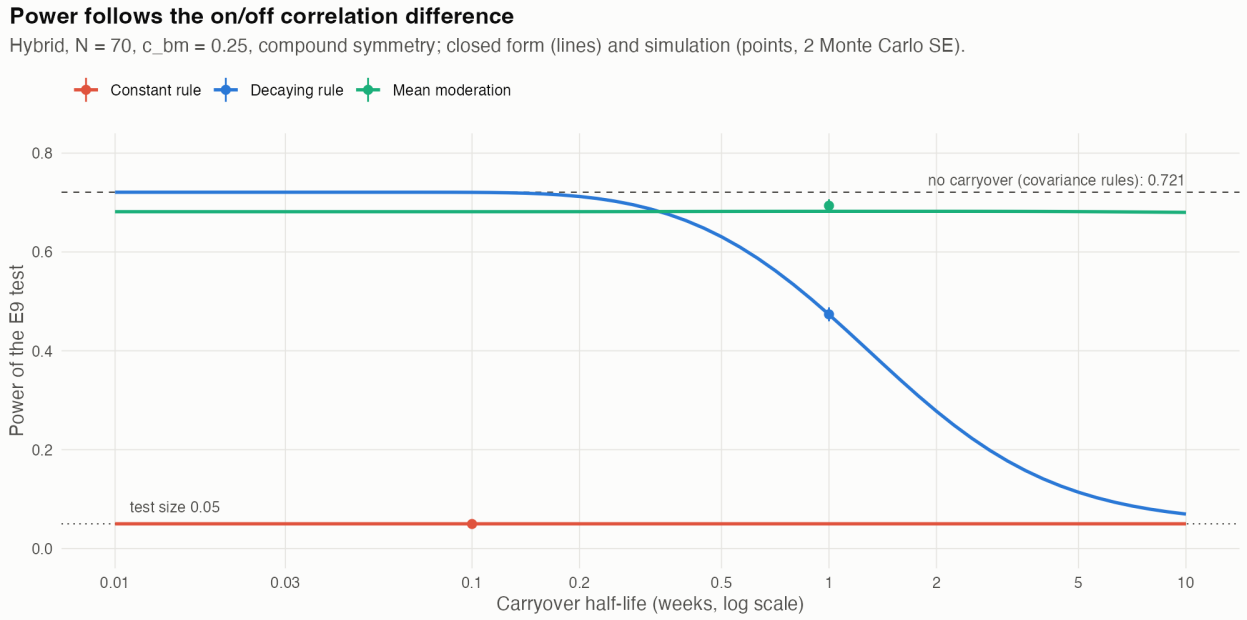


Figure 8: Closed-form E9 power by half-life under the three rules, with the simulated points.

Figure 8 completes the chain. Under the decaying rule power follows $\mathcal{G}$: flat through the pharmacological half-lives, then falling to 0.473 at one week and 0.278 at two. Under the constant rule it is the test size at every positive half-life. Under mean moderation it does not depend on the half-life.

12. Common misreadings

- **"Under the constant rule carryover weakens the interaction."** It removes it. The drug effect one week after stopping is 0.1% of its on-drug value at $t_{1/2} = 0.1$, yet the band is fully open. The loss is a property of the rule, not of carryover (paper 01, Stage 2).
- **"The band width is the drug effect."** It is not. The band is the difference the biomarker makes. The drug effect is the height of the trajectory above the off-drug trajectory; it can be large while the band is narrow, and under mean moderation off drug the mean can be positive while the band is closed (Figure 2, panel D).

- **"The published analysis kept some power, so the interaction was partly there."** That analysis counts baseline as an off-drug visit. At baseline $g = 0$, so baseline alone keeps a small difference: $\beta = -0.130/(n_{\text{off}} + 1)$ per path, pooled -0.029 and power 0.081 for the corresponding statistic E9b (`docs/37`, Section 5). Among the post-baseline visits there is no interaction.
- **"Equal expected slopes mean equal power."** Not across ways of generating the data (Table 6).

13. Exercises

1. Under the decaying rule with $t_{1/2} = 0.5$ week, compute $\bar{g}_{\text{off}}$, $\mathcal{G}$ and $\beta_{bm:D}$ for path 3.
2. With $c_{bm} = 0.20$ and no carryover, how wide is the band at an on-drug visit, and what is $\beta_{bm:D}$?
3. Explain in two sentences why the constant rule gives $\mathcal{G} = 1$ at $t_{1/2} = 0$ but $\mathcal{G} = 0$ at $t_{1/2} = 0.001$.
4. A colleague plans for 80% power at $N = 90$ assuming no carryover. Using Table 5, roughly how many participants are needed if the half-life is 0.5 week?
5. Suppose the biomarker were also correlated with the natural-course response TV at every visit. Would the band width change? Would the E9 slope change? (Hint: Section 4, and which visits TV contributes to.)

Answers. (1) The off-drug visits are at $t_{sd} = 1, 2, 3, 4$, so $g = 2^{-2}, 2^{-4}, 2^{-6}, 2^{-8} = 0.25, 0.0625, 0.0156, 0.0039$, $\bar{g}_{\text{off}} = 0.083$, $\mathcal{G} = 0.917$ and $\beta_{bm:D} = -0.130 \times 0.917 = -0.119$ (Table 5 gives 0.917 for path 3). (2) $W = 2 \times 0.20 \times 8 = 3.2$ points; $\beta_{bm:D} = -0.20 \times 8/15.36 = -0.104$. (3) At $t_{1/2} = 0$ the mean drug response off drug is exactly zero, so the constant rule sets $g = 0$ there; at any positive half-life the mean is positive, however small, so $g = 1$. (4) $90 \times 1.22 \approx 110$. (5) The total band would widen at every visit by the same amount, because TV enters on and off drug alike. If TV's correlation with the biomarker were the same at every visit, it would cancel from the on-minus-off contrast and leave the E9 slope unchanged; if it grew over the trial, as natural course does, it would not cancel, and the E9 slope would be biased. This is the prognostic-biomarker problem of paper 01, Stage 6, and `docs/48`.

14. Scope and limitations

- **Expectations hold for any correlation structure; variances do not.** Sections 3 to 9 use only the correlation of the biomarker with the drug response, so they hold under compound symmetry, under the separable AR(1) form of paper 01's recommended process, or any other valid structure. The variances, residual SDs and power of Sections 9 and 11 are for the compound-symmetry construct of the published process. Under separable AR(1) they differ (paper 01, Table rhostage).
- **E9 is a teaching statistic.** It has a closed form, which is why it is used here. Paper 01 plans with E9 and checks the result against the mixed-model analysis G10 by simulation; paper 02 recommends the analysis. Neither is derived here.
- **Feasibility.** At $c_{bm} = 0.25$ every correlation matrix in this tutorial is positive definite (the script checks), but under compound symmetry without carryover the bound is 0.256, close to 0.25. Larger values of c_{bm} are not feasible in that construct (`docs/34`, `docs/36`; paper 01, Stage 1).
- **The half-life of the correlation equals the carryover half-life** under the decaying rule as defined. If the biomarker's predictive value faded at a different rate from the drug's effect on the mean, the formulas still hold with that rate in g_t.
- **One effect size, one design.** All numbers are for the Hybrid design, $N = 70$ and $c_{bm} = 0.25$.

15. Reproducibility

```
Rscript analysis/scripts/quick-sim/hybrid-design-primer/\
03-band-width-tutorial.R
```

Run from the repository root. The script reads the published construct from `04-power-simulation.R` in `analysis/scripts/quick-sim/hendrickson-problems/`, writes Figures 1 to 8 to `docs/figures/76-fig*.png`,

and writes its tables to the directory `band-width/` in `analysis/data/quick-sim/hybrid-design-primer/`. The simulated trials are cached in `mc-e9-reps5000.csv` (seeds 7601 to 7604); delete the file to rerun them. The script stops unless it reproduces Table 9 of `docs/40`, the closed-form E9 power of `docs/37` (0.721, 0.050, 0.473, 0.682) and the simulation within 3 Monte Carlo SE.

16. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013. Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. Casella G, Berger RL. *Statistical Inference*, 2nd ed. Pacific Grove, CA: Duxbury; 2002. (The bivariate normal and its conditional distributions.)
3. pmsimstats team. `docs/37`: closed-form power of the paired-difference statistic under carryover.
4. pmsimstats team. `docs/40`: the Hendrickson Hybrid design primer.
5. pmsimstats team. Paper 01: `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`.